

CBSE Class 10 Mathematics — Coordinate Geometry

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions:

1. This question paper contains *questions. All questions are compulsory.*
1. Questions are grouped into sections A through E.
2. Use of calculators is NOT permitted.

Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. The distance of the point $P(3, 4)$ from the origin is:
(A) (A) 3
(B) (B) 4
(C) (C) 5
(D) (D) 7
2. The coordinates of the midpoint of the line segment joining the points $A(2, 4)$ and $B(4, 6)$ are:
(A) (A) (2, 3)
(B) (B) (3, 5)
(C) (C) (4, 4)
(D) (D) (6, 10)
3. The abscissa of the point (5, 8) is:
(A) (A) 5
(B) (B) 8
(C) (C) 3
(D) (D) 13
4. If the distance between points $(2, -2)$ and $(-1, x)$ is 5, then the value of x is not asked, but what is the distance formula expression?
(A) (A) $\sqrt{(2-1)^2 + (-2-x)^2}$
(B) (B) $\sqrt{(2+1)^2 + (-2+x)^2}$
(C) (C) $\sqrt{(1-2)^2 + (x-2)^2}$
(D) (D) $\sqrt{(-1-2)^2 + (x+2)^2}$

5. The point which lies on the y-axis is:
- (A) (A) (2, 0)
 - (B) (B) (1, 1)
 - (C) (C) (0, 5)
 - (D) (D) (5, 5)
6. Assertion (A): The point (0, 5) lies on the y-axis. Reason (R): For any point on the y-axis, the x-coordinate is zero.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 - (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 - (C) (C) Assertion (A) is true but Reason (R) is false
 - (D) (D) Assertion (A) is false but Reason (R) is true
7. Assertion (A): The distance of the point $P(3, 4)$ from the origin (0, 0) is 5 units. Reason (R): The distance of a point (x, y) from the origin is calculated by $\sqrt{x^2 + y^2}$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 - (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 - (C) (C) Assertion (A) is true but Reason (R) is false
 - (D) (D) Assertion (A) is false but Reason (R) is true
8. Assertion (A): The midpoint of the line segment joining (2, 4) and (6, 8) is (4, 6). Reason (R): The midpoint formula is $(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 - (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 - (C) (C) Assertion (A) is true but Reason (R) is false
 - (D) (D) Assertion (A) is false but Reason (R) is true
9. Assertion (A): The points (1, 1), (2, 2) and (3, 3) are collinear. Reason (R): Three points are collinear if the sum of distances between any two pairs is equal to the distance between the third pair.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 - (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 - (C) (C) Assertion (A) is true but Reason (R) is false

(D) (D) Assertion (A) is false but Reason (R) is true

10. Assertion (A): The area of a triangle with vertices $(0, 0)$, $(4, 0)$, and $(0, 3)$ is 6 sq units.
Reason (R): Area of a triangle with coordinates $(0, 0)$, (x_1, y_1) , (x_2, y_2) is $\frac{1}{2}|x_1y_2 - x_2y_1|$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
- (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
- (C) (C) Assertion (A) is true but Reason (R) is false
- (D) (D) Assertion (A) is false but Reason (R) is true

Section D: Long Answer (4-5 marks each)

1. Find the ratio in which the line segment joining the points $A(-3, 10)$ and $B(6, -8)$ is divided by the point $P(-1, 6)$.
2. If the points $A(6, 1)$, $B(8, 2)$, $C(9, 4)$, and $D(p, 3)$ are the vertices of a parallelogram taken in order, find the value of p .
3. Find the distance between the points $P(a + b, b - a)$ and $Q(a - b, a + b)$.
4. Find the value of y for which the distance between the points $P(2, -3)$ and $Q(10, y)$ is 10 units.
5. Find the coordinates of a point A , where AB is the diameter of a circle whose centre is $(2, -3)$ and B is $(1, 4)$.
6. Find the ratio in which the point $P(x, 2)$ divides the line segment joining the points $A(12, 5)$ and $B(4, -3)$. Also, find the value of x .
7. If the points $A(6, 1)$, $B(8, 2)$, $C(9, 4)$, and $D(p, 3)$ are the vertices of a parallelogram taken in order, find the value of p .
8. Find the area of a quadrilateral whose vertices, taken in order, are $(-4, -2)$, $(-3, -5)$, $(3, -2)$, and $(2, 3)$.
9. Find the coordinates of the points which divide the line segment joining $A(-2, 2)$ and $B(2, 8)$ into four equal parts.